

InGen

1.-la distribución que creo que es la mejor para el proyecto Ubuntu Server, por estabilidad, soporte y comunidad

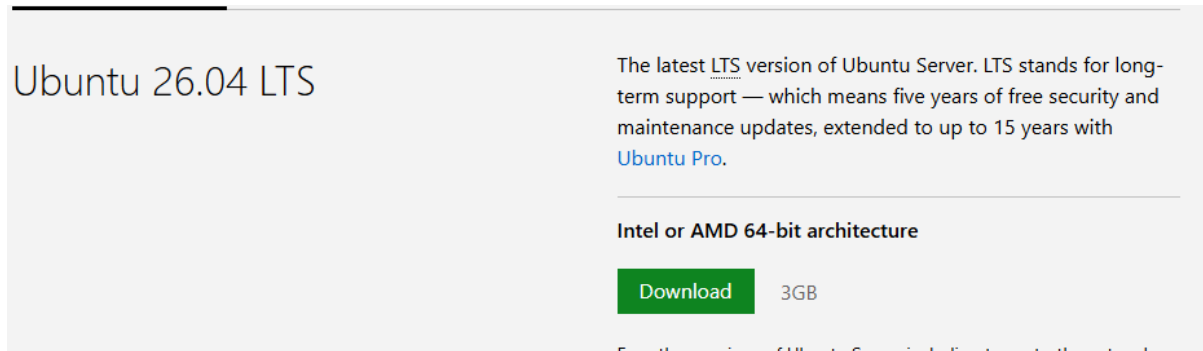
-comparación con Windows

Ubuntu: modular, libre, seguro y escalable

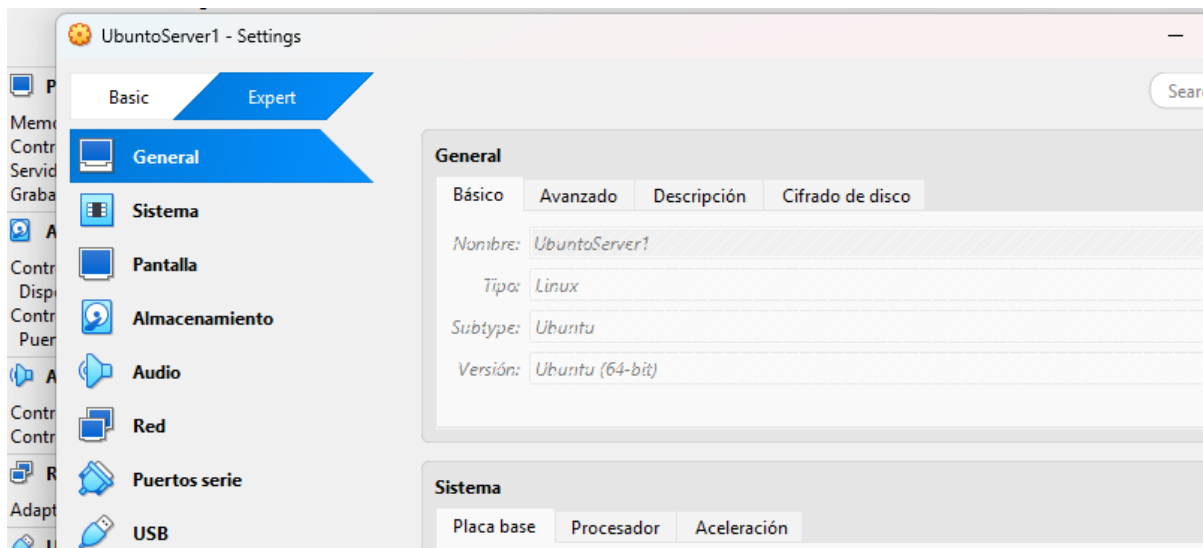
Windows: más orientado a GUI, licencias y menos flexible en servidores

2.-

-descargar Ubuntu server desde página oficial la instalación va a ser en una máquina virtual (VirtualBox), así que no es necesario el uso de herramientas como rufus para convertir memorias USB booteables



Configuración de parámetros en maquina en VirtualBox



Después de la instalación crear usuario Root

```

isak@UbuntuServer:~$ sudo adduser admin
[sudo: authenticate] Password:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for admin
Enter the new value, or press ENTER for the default
    Full Name []: isaac
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
isak@UbuntuServer:~$ sudo usermod -aG sudo admin
isak@UbuntuServer:~$ groups admin
admin : admin sudo users
isak@UbuntuServer:~$ su - admin
Password:
admin@UbuntuServer:~$

```

Configuración de red básica

```

# This is the network config written by 'subiquity'
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s3:
      dhcp4: no
      dhcp6: no
      addresses:
        - 192.168.1.50/24
      routes:
        - to: default
          via: 192.168.1.1
      nameservers:
        addresses: [8.8.8.8, 1.1.1.1]

```

Habilitación de SSH en servidor

```

admin@UbuntuServer:~$ sudo apt install openssh-server
Installing:
  openssh-server

Installing dependencies:
  libwrap0  ncurses-term  openssh-sftp-server  ssh-import-id

Suggested packages:
  molly-guard  monkeysphere  ssh-askpass

Summary:
  Upgrading: 0, Installing: 5, Removing: 0, Not Upgrading: 20
  Download size: 998 kB
  Space needed: 7,690 kB / 46.1 GB available

Continue? [Y/n] y

```

Iniciar servicio ssh con `systemctl start ssh`, ver estado del servicio con `systemctl status ssh` y habilitación desde el arranque con `systemctl enable ssh`

```

admin@UbuntuServer:~$ sudo systemctl start ssh
admin@UbuntuServer:~$ systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Fri 2026-06-26 16:28:44 UTC; 2s ago
     Invocation: 58d6e5befdd843dfa0b13f63b31abf4c
   TriggeredBy: * ssh.socket
      Docs: man:sshd(8)
            man:sshd_config(5)
    Process: 3569 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 3572 (sshd)
      Tasks: 1 (limit: 8459)
     Memory: 1.9M (peak: 2.2M)
        CPU: 113ms
    CGroup: /system.slice/ssh.service
            └─3572 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

admin@UbuntuServer:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink '/etc/systemd/system/sshd.service' → '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
admin@UbuntuServer:~$

```

cambio de puerto de ssh por seguridad y deshabilitar acceso directo de root en archivo de configuración `/etc/ssh/sshd_config`

```

#
Port 2222
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

```

Prueba de conexión desde pc físico

```

The authenticity of host '192.168.1.50 (192.168.1.50)' can't be established.
ED25519 key fingerprint is SHA256:R1Yi/+UErbn2kx1DVy6Jky+uXujFTHFqMLBp3fmmmao.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.50' (ED25519) to the list of known hosts.
admin@192.168.1.50's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-22-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Jun 26 04:41:57 PM UTC 2026

```

```

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

```

```

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

```

```

admin@UbuntuServer:~$ pwd
/home/admin
admin@UbuntuServer:~$

```

3.-crea estructura resquema de carpetas bases del servidor

```

admin@UbuntuServer:~$ sudo mkdir -p /srv/www
[sudo: authenticate] Password:
admin@UbuntuServer:~$ sudo mkdir -p /backups
admin@UbuntuServer:~$ sudo mkdir -p /usuarios
admin@UbuntuServer:~$ sudo mkdir -p /srv/scripts
admin@UbuntuServer:~$ sudo mkdir -p /srv/logs
admin@UbuntuServer:~$ _

```

El parámetro -p crea las carpetas necesarias sin error si ya existen

Comandos de creación y copia

```

admin@UbuntuServer:~$ #crear archivo
admin@UbuntuServer:~$ sudo touch /srv/www/index.html
admin@UbuntuServer:~$ #copiar archivo
admin@UbuntuServer:~$ ls -l /srv/www
total 0
-rw-r--r-- 1 root root 0 Jun 26 19:30 index.html
admin@UbuntuServer:~$ sudo cp /srv/www/index.html /backups/
admin@UbuntuServer:~$ ls -l /backups/
total 0
-rw-r--r-- 1 root root 0 Jun 26 19:31 index.html

```

Movimiento y eliminación

```

admin@UbuntuServer:~$ #crear archivo
admin@UbuntuServer:~$ sudo touch /srv/www/index.html
admin@UbuntuServer:~$ #copiar archivo
admin@UbuntuServer:~$ ls -l /srv/www
total 0
-rw-r--r-- 1 root root 0 Jun 26 19:30 index.html
admin@UbuntuServer:~$ sudo cp /srv/www/index.html /backups/
admin@UbuntuServer:~$ ls -l /backups/
total 0
-rw-r--r-- 1 root root 0 Jun 26 19:31 index.html

```

Movimiento y eliminación

```

admin@UbuntuServer:~$ sudo mv /srv/www/index.html /srv/www/inicio.html
admin@UbuntuServer:~$ sudo rm /srv/www/inicio.html

```

Validar la eliminación del archivo

```
admin@UbuntuServer:~$ sudo find /srv/www -name inicio.html
admin@UbuntuServer:~$ _
```

Enlaces simbólicos

Un enlace simbólico es como un atajo: apunta a otro archivo o carpeta.

Cuando accedes al enlace, el sistema te lleva automáticamente al destino real.

Ejemplo:

```
admin@UbuntuServer:~$ sudo ln -s /srv/www /home/admin/www_link
admin@UbuntuServer:~$
```

crea un enlace llamado `www_link` dentro de la carpeta personal del usuario `admin`
ese enlace apunta a `/srv/www`

```
admin@UbuntuServer:~$ ls -l /home/admin/
total 0
lrwxrwxrwx 1 root root 8 Jun 26 19:41 www_link -> /srv/www
admin@UbuntuServer:~$ _
```

Modificar permisos y propietarios

```
admin@UbuntuServer:~$ sudo chmod 770 /srv/www
admin@UbuntuServer:~$ sudo chmod 750 /backups
admin@UbuntuServer:~$ ls -ld /srv/www
drwxrwx--- 2 root root 4096 Jun 26 19:36 /srv/www
admin@UbuntuServer:~$ ls -ld /backups
drwxr-x--- 2 root root 4096 Jun 26 19:31 /backups
admin@UbuntuServer:~$ _
```

Configurar umask (permisos por defecto)

esto evita que nuevos archivos sean legibles por otros usuarios

```
-----
# /etc/profile: system-wide .profile file for the Bourne shell (sh(1))
# and Bourne compatible shells (bash(1), ksh(1), ash(1), ...).

if [ "${PS1-}" ]; then
  if [ "${BASH-}" ] && [ "$BASH" != "/bin/sh" ]; then
    # The file bash.bashrc already sets the default PS1.
    # PS1='\h:\w\$ '
    if [ -f /etc/bash.bashrc ]; then
      . /etc/bash.bashrc
    fi
  else
    if [ "$(id -u)" -eq 0 ]; then
      PS1='# '
    else
      PS1='~$ '
    fi
  fi
fi

if [ -d /etc/profile.d ]; then
  for i in $(run-parts --list --regex '^[a-zA-Z0-9_][a-zA-Z0-9._-]*\.sh$' /etc/profile.d); do
    if [ -r $i ]; then
      . $i
    fi
  done
unset i
fi

umask 027
-----
```

Documentar jerarquía visual

con el comando tree se puede generar un esquema

```
admin@UbuntuServer:~$ tree /srv /backups /usuarios
/srv
├── logs
├── scripts
└── www [error opening dir]
/backups [error opening dir]
/usuarios

4 directories, 1 file
admin@UbuntuServer:~$ _
```

4.-

Creación de usuarios y grupos

```
root@UbuntuServer:/home/admin# sudo groupadd desarrolladores
root@UbuntuServer:/home/admin# sudo groupadd gas
root@UbuntuServer:/home/admin# sudo groupadd tecnicos
root@UbuntuServer:/home/admin#
root@UbuntuServer:/home/admin# #creacion usuarios
root@UbuntuServer:/home/admin# sudo useradd -m -s /bin/bash des1 -g desarrolladores
root@UbuntuServer:/home/admin# sudo useradd -m -s /bin/bash cua1 -g gas
root@UbuntuServer:/home/admin# sudo useradd -m -s /bin/bash tec1 -g tecnicos
root@UbuntuServer:/home/admin# _
```

asignar permisos sobre carpetas criticas

```
root@UbuntuServer:/home/admin# sudo chown -R des1:desarrolladores /srv/www
root@UbuntuServer:/home/admin# sudo chmod 770 /srv/www
root@UbuntuServer:/home/admin# _
```

```
root@UbuntuServer:/home/admin# sudo chown -R cua1:gas /backups
root@UbuntuServer:/home/admin# sudo chmod 750 /backups
root@UbuntuServer:/home/admin#
```

configurar sudo restringido

```
# Allow members of group sudo to execute any command
%sudo    ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@includedir /etc/sudoers.d

tec1     ALL=(ALL) /bin/systemctl restart ssh, /bin/systemctl status ssh
```

⌘G Help ⌘O Write Out ⌘F Where Is ⌘K Cut ⌘T Execute ⌘C Locati

tabla de usuarios y grupos

```

root@UbuntuServer:/home/admin# id des1
uid=1007(des1) gid=1005(desarrolladores) groups=1005(desarrolladores)
root@UbuntuServer:/home/admin# id cua1
uid=1008(cua1) gid=1006(qas) groups=1006(qas)
root@UbuntuServer:/home/admin# id tec1
uid=1009(tec1) gid=1007(tecnicos) groups=1007(tecnicos)
root@UbuntuServer:/home/admin# _

```

usuario	UID	grupo	GID	ROL	Permisos asignados
des1	1007	Qas	1005	Desarrollo web	Rwx em /srv/www
Cua	1008	qas	1006	Control de calidad	r-x en backups
tec	1009	tecnicos	1004	Soporte tecnico	Sudo limitado (reinicio y estadl de SSH)

5.-

gestión de procesos

Utilizar comandos ps, top, kill, nice, cron para gestionar servicios.

Ejemplos con el servicio SSH

```

root@UbuntuServer:/home/admin# ps aux | grep ssh
root      4095  0.0  0.0 10744 7888 ?        Ss   16:41   0:00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
root      5409  0.0  0.0  6716  2644 pts/0    S+   22:01   0:00 grep --color=auto ssh
root@UbuntuServer:/home/admin# _

```

Monitoreo con top

```

top - 22:02:38 up 6:31, 1 user, load average: 0.00, 0.01, 0.00
Tasks: 147 total, 1 running, 146 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.2 sy, 0.0 ni, 99.4 id, 0.0 wa, 0.0 hi, 0.3 si, 0.0 st
MiB Mem : 9428.1 total, 8209.5 free, 512.5 used, 963.0 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 8915.7 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
  5412 root        20   0   10828    6348   4184 R   0.7   0.1   0:00.05 top
  5131 root        20   0   12400    6440   5672 S   0.3   0.1   0:00.55 sudo
    1 root        20   0   25824   16832  11560 S   0.0   0.2   0:11.20 systemd
    2 root        20   0         0         0      0 S   0.0   0.0   0:00.39 kthreadd
    3 root        20   0         0         0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
    4 root         0 -20         0         0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
    5 root         0 -20         0         0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
    6 root         0 -20         0         0      0 I   0.0   0.0   0:00.00 kworker/R-kufree_rcu_reclaim
    7 root         0 -20         0         0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
    8 root         0 -20         0         0      0 I   0.0   0.0   0:00.00 kworker/R-netns

```

Detener servicio con kill

```

root@UbuntuServer:/home/admin# ps aux | grep sleep
root      5428  0.0  0.0 16112 7748 pts/0    S   22:06   0:00  sleep 300
root      5432  0.0  0.0  6716 2688 pts/0    S+  22:07   0:00  grep --color=auto sleep
root@UbuntuServer:/home/admin# kill -9 5428
root@UbuntuServer:/home/admin# ps aux | grep sleep
root      5434  0.0  0.0  6716 2688 pts/0    S+  22:08   0:00  grep --color=auto sleep
[1]+  Killed                  sleep 300
root@UbuntuServer:/home/admin# _

```

Nice

```

root@UbuntuServer:/home/admin# nice -n 10 sleep 300 &
[1] 5447
root@UbuntuServer:/home/admin# _

```

Cron

```

#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow  command
* * * * * echo "cron ejecutado a $(date)" >> /home/admin/cron_prueba.log

```

archivo creado con cron

```

root@UbuntuServer:/home/admin# sudo systemctl start cron
root@UbuntuServer:/home/admin# cat /home/admin/cron_prueba.log
cron ejecutado a Fri Jun 26 10:17:01 PM UTC 2026
cron ejecutado a Fri Jun 26 10:18:01 PM UTC 2026
root@UbuntuServer:/home/admin# _

```

Crear una tarea programada con cron para realizar backups diarios en /backups.

Creación pequeño script para que realice el backup diario en /usr/local/bin/backup_diario.sh

```
GNU nano 8.7.1 /usr/local/bin/backup_diario.sh *
#!/bin/bash
# script de respaldo diario
FECHA=$(date +%Y-%m-%d_%H-%M)
cp -r /srv/www /backups/respaldo_$FECHA
echo "Backup realizado el $FECHA" >> /backups/registro_backups.log_
```

entregarle permisos de ejecución

```
Archivo Máquina Ver Entrada Dispositivos Ayuda
root@UbuntuServer:/home/admin# sudo chmod +x /usr/local/bin/backup_diario.sh
root@UbuntuServer:/home/admin#
```

en cron agregar linea para que ejecute el script y realice el respaldo todos los días a las 2 am

```
GNU nano 8.7.1 /ti
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow command
0 2 * * * /usr/local/bin/backup_diario.sh_
```

validar estado del servicio

```
root@UbuntuServer:/home/admin# sudo systemctl status cron
* cron.service - Regular background program processing daemon
   Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-06-26 15:30:50 UTC; 7h ago
  Invocation: ecbb27ad6e942b0acce935d77511d96
     Docs: man:cron(8)
    Main PID: 1182 (cron)
      Tasks: 1 (limit: 8459)
     Memory: 516K (peak: 4.2M)
        CPU: 731ms
    CGroup: /system.slice/cron.service
            └─1182 /usr/sbin/cron -f -P
```

prueba de ejecución del servicio

```
root@UbuntuServer:/home/admin# sudo /usr/local/bin/backup_diario.sh
root@UbuntuServer:/home/admin# ls /backups
index.html  registro_backups.log  respaldo_2026-06-26_22-33
```

```
root@UbuntuServer:/home/admin# car /backups/registro_backups.log
passphrase:
confirm:
-----BEGIN CAR STREAM-----
6Iz7Hs932dUKcaMN6Hn1CjyhH6wT0s91X1fIQ0iZoHig0UnX/Q75UOK1o7o/N7I9
LxauX0kgFUHy1BUufVIP3NIaA3KwYeSSrHQ6X2B1XpNuPeTqbImEIPcSYIC4fq/7
-----END CAR STREAM-----
root@UbuntuServer:/home/admin# cat /backups/registro_backups.log
Backup realizado el 2026-06-26_22-33
root@UbuntuServer:/home/admin#
```

Visualizar demonios activos con systemctl y detener/iniciar uno con logs.

Ver todos los demonios activos

```
root@UbuntuServer:/home/admin# systemctl list-units --type=service --state=running
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
chrony.service                     loaded active running chrony, an NTP client/server
cron.service                       loaded active running Regular background program processing daemon
dbus.service                       loaded active running D-Bus System Message Bus
fwupd.service                      loaded active running Firmware update daemon
getty@tty1.service                 loaded active running Getty on tty1
ModemManager.service              loaded active running Modem Manager
multipathd.service                loaded active running Device-Mapper Multipath Device Controller
networkd-dispatcher.service        loaded active running Dispatcher daemon for systemd-networkd
polkit.service                     loaded active running Authorization Manager
rsyslog.service                   loaded active running System Logging Service
ssh.service                        loaded active running OpenBSD Secure Shell server
systemd-journald.service           loaded active running Journal Service
systemd-logind.service             loaded active running User Login Management
systemd-networkd.service           loaded active running Network Management
systemd-resolved.service           loaded active running Network Name Resolution
systemd-udevd.service              loaded active running Rule-based Manager for Device Events and Files
udisks2.service                   loaded active running Disk Manager
unattended-upgrades.service         loaded active running Unattended Upgrades Shutdown
upower.service                     loaded active running Daemon for power management
user@1000.service                  loaded active running User Manager for UID 1000

Legend: LOAD    → Reflects whether the unit definition was properly loaded.
          ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
          SUB    → The low-level unit activation state, values depend on unit type.

20 loaded units listed.
```

verificar estado de un servicio

```
root@UbuntuServer:/home/admin# sudo systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-06-26 16:41:03 UTC; 6h ago
     Invocation: 2bc75d3cefb849a2ad777f1c64f2e122
   TriggeredBy: * ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Process: 4092 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 4095 (sshd)
     Tasks: 1 (limit: 8459)
    Memory: 1.2M (peak: 21M)
```

detener y activar servicio

```

root@UbuntuServer:/home/admin# sudo systemctl stop ssh
Stopping 'ssh.service', but its triggering units are still active:
ssh.socket
root@UbuntuServer:/home/admin# sudo systemctl start ssh
root@UbuntuServer:/home/admin#

```

ver logs del servicio

```

root@UbuntuServer:/home/admin# sudo journalctl -u ssh -n 5
Jun 26 22:56:09 UbuntuServer systemd[1]: Stopped ssh.service - OpenBSD Secure Shell server.
Jun 26 22:56:23 UbuntuServer systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jun 26 22:56:23 UbuntuServer sshd[5801]: Server listening on 0.0.0.0 port 22.
Jun 26 22:56:23 UbuntuServer sshd[5801]: Server listening on :: port 22.
Jun 26 22:56:23 UbuntuServer systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
root@UbuntuServer:/home/admin#

```

ver logs en tiempo real

```

root@UbuntuServer:/home/admin# sudo journalctl -u ssh -f
Jun 26 16:41:57 UbuntuServer sshd-session[4128]: Accepted password for admin from 192.168.1.26 port 6
Jun 26 16:41:57 UbuntuServer sshd-session[4128]: pam_unix(sshd:session): session opened for user admin
Jun 26 22:56:09 UbuntuServer systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server...
Jun 26 22:56:09 UbuntuServer sshd[4095]: Received signal 15; terminating.
Jun 26 22:56:09 UbuntuServer systemd[1]: ssh.service: Deactivated successfully.
Jun 26 22:56:09 UbuntuServer systemd[1]: Stopped ssh.service - OpenBSD Secure Shell server.
Jun 26 22:56:23 UbuntuServer systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jun 26 22:56:23 UbuntuServer sshd[5801]: Server listening on 0.0.0.0 port 22.
Jun 26 22:56:23 UbuntuServer sshd[5801]: Server listening on :: port 22.
Jun 26 22:56:23 UbuntuServer systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

```

Monitorear consumo de recursos

con top

```

top - 00:14:02 up 8:42, 1 user, load average: 0.07, 0.05, 0.01
Tasks: 147 total, 1 running, 146 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.1 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 9428.1 total, 8199.4 free, 516.2 used, 969.6 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 8912.0 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
 1081 message+  20   0   9156   5852   4724 S   0.3   0.1   0:24.91 dbus-daemon
 2308 root        20   0 289080   3908   3456 S   0.3   0.0   0:37.88 VBoxService
 6146 root        20   0     0     0     0 I   0.3   0.0   0:07.58 kworker/0:1-events
 6245 root        20   0  10828   6352   4192 R   0.3   0.1   0:00.05 top
    1 root        20   0 25824  16832  11560 S   0.0   0.2   0:12.57 systemd
    2 root        20   0     0     0     0 S   0.0   0.0   0:00.47 kthreadd
    3 root        20   0     0     0     0 S   0.0   0.0   0:00.00 pool_workqueue_release

```

con ps

```

root@UbuntuServer:/home/admin# ps -eo pid,comm,%cpu,%mem --sort=%cpu | head
  PID COMMAND           %CPU %MEM
    3 pool_workqueue_     0.0  0.0
    4 kworker/R-rcu_g      0.0  0.0
    5 kworker/R-sync_     0.0  0.0
    6 kworker/R-kofre     0.0  0.0
    7 kworker/R-slub_     0.0  0.0
    8 kworker/R-netns     0.0  0.0
   12 kworker/u16:0-i     0.0  0.0
   13 kworker/R-mm_pe     0.0  0.0
   16 rcu_exp_par_gp_     0.0  0.0

```

con vmstat y iostat

```

root@UbuntuServer:/home/admin# vmstat
procs-----memory-----swap-----io-----system-----cpu-----
 r b swpd free buff cache si so bi bo in cs us sy id wa st gu
 1 0 0 8398404 41660 951404 0 0 24 15 197 0 0 0 100 0 0 0 0
root@UbuntuServer:/home/admin# iostat -xz 5
Linux 7.0.0-22-generic (UbuntuServer) 06/27/2026 _x86_64_ (4 CPU)

avg-cpu:  %user   %nice    %system %iowait  %steal   %idle
           0.04    0.00     0.35    0.09    0.00   99.53

Device:            r/s     kB/s    rrqm/s   %rrqm r_await rareq-sz    u/s    kB/s    wrqm/s   %wrqm w_await wareq-sz    d/s    kB/s    drqm/s   %drqm d_await
loop0              0.00      0.00      0.00    0.00  0.00  0.10  1.00  0.00    0.00    0.00  0.00  0.00  0.00    0.00    0.00    0.00  0.00
loop1              0.00      0.00      0.00    0.00  0.00  0.00  0.00  0.00    0.00    0.00  0.00  0.00  0.00    0.00    0.00    0.00  0.00
sda                0.73    24.43    0.29   28.69  11.37  33.43  0.50   15.58    0.25   33.30  7.29  31.02  0.00    0.00    0.00    0.00  0.00
sdb                0.00      0.00      0.00    0.00  0.00  0.00  0.00    0.00    0.00    0.00  0.00  0.00  0.00    0.00    0.00    0.00  0.00
sr0                0.00      0.00      0.00    0.00  0.00  0.00  0.00    0.00    0.00    0.00  0.00  0.00  0.00    0.00    0.00    0.00  0.00

avg-cpu:  %user   %nice    %system %iowait  %steal   %idle
           0.05    0.00     0.33    0.05    0.00   99.57

Device:            r/s     kB/s    rrqm/s   %rrqm r_await rareq-sz    u/s    kB/s    wrqm/s   %wrqm w_await wareq-sz    d/s    kB/s    drqm/s   %drqm d_await
loop0              0.00      0.00      0.00    0.00  0.00  0.00  0.00    0.00    0.00    0.00  0.00  0.00  0.00    0.00    0.00    0.00  0.00
loop1              0.00      0.00      0.00    0.00  0.00  0.00  0.00    0.00    0.00    0.00  0.00  0.00  0.00    0.00    0.00    0.00  0.00
sda                0.39     2.34    0.19   33.33  3.00  6.00  0.00    0.00    0.00    0.00  0.00  0.00  0.00    0.00    0.00    0.00  0.00
sdb                0.00      0.00      0.00    0.00  0.00  0.00  0.00    0.00    0.00    0.00  0.00  0.00  0.00    0.00    0.00    0.00  0.00
sr0                0.00      0.00      0.00    0.00  0.00  0.00  0.00    0.00    0.00    0.00  0.00  0.00  0.00    0.00    0.00    0.00  0.00

```

6.-

Servicios de red en Linux

Instalar Apache (HTTP), OpenSSH, vsftpd (FTP) y DHCP.

Instalar apache (HTTP)

```

root@UbuntuServer:/home/admin# sudo apt install apache2 -y
apache2 is already the newest version (2.4.66-2ubuntu2.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 20
root@UbuntuServer:/home/admin# _

```

verificar estado

```

root@UbuntuServer:/home/admin# sudo systemctl status apache2
* apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-06-27 00:31:17 UTC; 3min 26s ago
   Invocation: 33675100d1db400b819ccb42a2a60798
   Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 7053 (apache2)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
   Tasks: 55 (limit: 8459)
   Memory: 6.1M (peak: 6.3M)
   CPU: 376ms
   CGroup: /system.slice/apache2.service

```

instalar open ssh

```

root@UbuntuServer:/home/admin# sudo apt install openssh-server -y
openssh-server is already the newest version (1:10.2p1-2ubuntu3.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 20
root@UbuntuServer:/home/admin# _

```

ver estado del servicio

```

root@UbuntuServer:/home/admin# sudo systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-06-26 22:56:23 UTC; 1h 40min ago
 Invocation: 5ea9860692234bd2b490af493ec0785a
 TriggeredBy: * ssh.socket
    Docs: man:sshd(8)
          man:sshd_config(5)
   Main PID: 5801 (sshd)
     Tasks: 1 (limit: 8459)

```

instalar vsftpd (FTP)

```

root@UbuntuServer:/home/admin# sudo apt install vsftpd -y
Installing:
  vsftpd

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 20
  Download size: 122 kB
  Space needed: 316 kB / 46.1 GB available

Get:1 http://cl.archive.ubuntu.com/ubuntu resolute/main amd64 vsftpd
Fetched 122 kB in 0s (314 kB/s)
Preconfiguring packages ...

```

ver estado del servicio

```

root@UbuntuServer:/home/admin# sudo systemctl status vsftpd
* vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-06-27 00:38:23 UTC; 56s ago
 Invocation: 22e615883849449686b343469253ebee
   Process: 7683 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SUCCESS)
   Main PID: 7685 (vsftpd)
     Tasks: 1 (limit: 8459)
    Memory: 872K (peak: 1.9M)
       CPU: 57ms

```

configuración básica en archivo /etc/vsftpd.conf

```

#
# Allow anonymous FTP? (Disabled by default).
anonymous_enable=NO
#
# Uncomment this to allow local users to log in.
local_enable=YES
#
# Uncomment this to enable any form of FTP write command.
write_enable=YES
#
# Default umask for local users is 077. You may wish to change
# if your users expect that (022 is used by most other ftpd's.
#local_umask=022
#

```

instalar DHCP

```

root@UbuntuServer:/home/admin# sudo apt install isc-dhcp-server -y
Installing:
  isc-dhcp-server

Installing dependencies:
  isc-dhcp-common

Suggested packages:
  isc-dhcp-server-ldap  policycoreutils

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 20
  Download size: 1,281 kB
  Space needed: 4,281 kB / 46.1 GB available

```

editar configuración en archivo /etc/dhcp/dhcpd.conf

```

root@UbuntuServer:/home/admin# sudo dhcpd -t
Internet Systems Consortium DHCP Server 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
root@UbuntuServer:/home/admin#

```

verificar estado del servicio

```

root@UbuntuServer:/home/admin# sudo systemctl status isc-dhcp-server
* isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-06-27 00:58:59 UTC; 14s ago
   Invocation: 02984cb406c5446d941b2a903233dc29
     Docs: man:dhcpd(8)
   Main PID: 8279 (dhcpd)
     Tasks: 1 (limit: 8459)
    Memory: 4M (peak: 4.2M)

```

habilitar servicio para que tenga inicio automático al arrancar sistema

```

root@UbuntuServer:/home/admin# sudo systemctl enable isc-dhcp-server
Synchronizing state of isc-dhcp-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable isc-dhcp-server
root@UbuntuServer:/home/admin#

```

Probar el acceso desde una segunda máquina (cliente) vía

navegador, FTP o SSH.

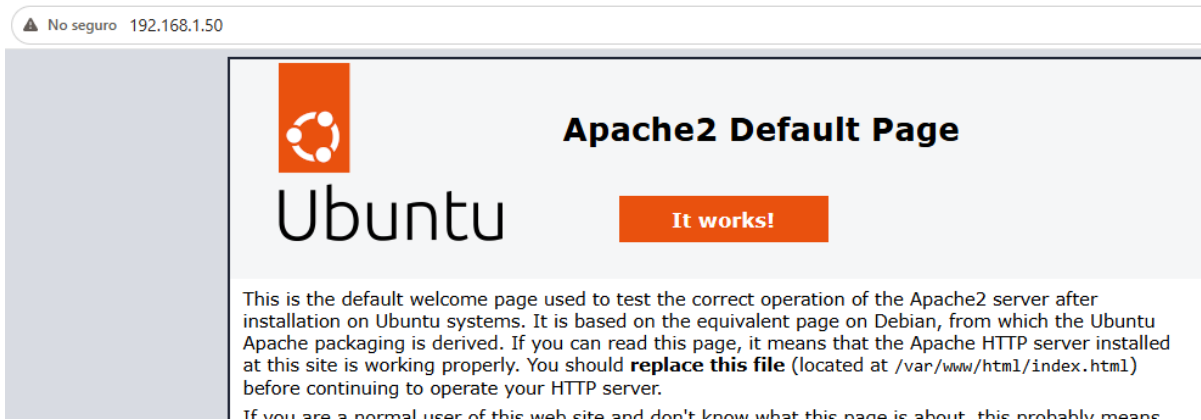
Ping desde equipo físico

```
C:\Users\PC Gamer>ping 192.168.1.50

Haciendo ping a 192.168.1.50 con 32 bytes de datos:
Respuesta desde 192.168.1.50: bytes=32 tiempo=1ms TTL=64
Respuesta desde 192.168.1.50: bytes=32 tiempo=1ms TTL=64
Respuesta desde 192.168.1.50: bytes=32 tiempo=1ms TTL=64
Respuesta desde 192.168.1.50: bytes=32 tiempo=3ms TTL=64

Estadísticas de ping para 192.168.1.50:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
        (0% perdidos),
    Tiempos aproximados de ida y vuelta en milisegundos:
```

Acceso vía navegador



Acceso desde SSH

```

C:\Users\PC Gamer>ssh isak@192.168.1.50
isak@192.168.1.50's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-22-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Jun 27 01:04:31 AM UTC 2026

System load:            0.01
Usage of /:             7.2% of 48.91GB
Memory usage:          3%
Swap usage:            0%
Processes:             160
Users logged in:       0
IPv4 address for enp0s3: 192.168.1.50
IPv6 address for enp0s3: 2803:c600:d209:8cf3:a00:27ff:feeb:4aaf

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

27 updates can be applied immediately.
15 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

isak@UbuntuServer:~$ pwd
/home/isak

```

7.-

Administración del almacenamiento en Linux

identificar dispositivos

Lsblk

```

root@UbuntuServer:/home/admin# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0   50G  0 disk
├─sda1       8:1    0    1M  0 part
└─sda2       8:2    0   50G  0 part /
sr0         11:0    1 58.5M  0 rom
root@UbuntuServer:/home/admin#

```

fdisk -l

```

root@UbuntuServer:/home/admin# sudo fdisk -l
Disk /dev/sda: 50 GiB, 53687091200 bytes, 104857600 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: D67F19BB-68DE-4221-B5AE-0756E0E1C0F1

Device        Start      End   Sectors  Size  Type
/dev/sda1     2048      4095    2048     1M  BIOS boot
/dev/sda2     4096 104855551 104851456 50G  Linux filesystem
root@UbuntuServer:/home/admin# _

```

parted -l

```

root@UbuntuServer:/home/admin# parted -l
Model: ATA VBOX HARDDISK (scsi)
Disk /dev/sda: 53.7GB
Sector size (logical/physical): 512B/512B
Partition Table: gpt
Disk Flags:

Number  Start   End     Size    File system  Name  Flags
  1      1049kB  2097kB  1049kB             bios_grub
  2      2097kB  53.7GB  53.7GB  ext4

Warning: Unable to open /dev/sr0 read-write (Read-only file system). /dev/sr0
has been opened read-only.
Error: /dev/sr0: unrecognised disk label
Model: VBOX CD-ROM (scsi)
Disk /dev/sr0: 61.4MB
Sector size (logical/physical): 2048B/2048B
Partition Table: unknown
Disk Flags:

```

Crear partición para /srv

```

Disk /dev/sdb: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb4abacb7

Command (m for help): n
Partition type
  p   primary (0 primary, 0 extended, 4 free)
  e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-20971519, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-20971519, default 20971519): +5G
Created a new partition 1 of type 'Linux' and of size 5 GiB.

```

/backups

```

Created a new partition 1 of type 'Linux' and of size 5 GiB.

Command (m for help): n
Partition type
   p   primary (1 primary, 0 extended, 3 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (2-4, default 2): 2
First sector (10487808-20971519, default 10487808):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (10487808-20971519, default 20971519):

Created a new partition 2 of type 'Linux' and of size 5 GiB.

```

formatear particiones

```

isak@UbuntuServer:~$ sudo mkfs.ext4 -L srv /dev/sdb1
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 1310720 4k blocks and 327680 inodes
Filesystem UUID: 07739997-1262-4aca-9d0a-9402cbd36745
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736

```

```

isak@UbuntuServer:~$ sudo mkfs.ext4 -L backups /dev/sdb2
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 1310464 4k blocks and 327680 inodes
Filesystem UUID: 38eb7923-718c-4041-a277-cc1f8e0e8b11
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736

```

montar directorios

```

isak@UbuntuServer:~$ sudo mount /dev/sdb1 /srv
isak@UbuntuServer:~$ sudo mount /dev/sdb2 /backups

```

Implementar un LVM que permita ampliar dinámicamente el espacio de /backups.

```

isak@UbuntuServer:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda   8:0    0  50G  0 disk
├─sda1 8:1    0   1M  0 part
├─sda2 8:2    0  50G  0 part /
└─sdb   8:16   0  10G  0 disk
   ├─sdb1 8:17   0   5G  0 part /srv
   └─sdb2 8:18   0   5G  0 part /backups
sr0   11:0    1 58.5M  0 rom

isak@UbuntuServer:~$ sudo blkid
[sudo: authenticate] Password:
/dev/sdb2: LABEL="backups" UUID="38eb7923-718c-4041-a277-cc1f8e0e8b11" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="b4abacb7-02"
/dev/sdb1: LABEL="srv" UUID="07739997-1262-4aca-9d0a-9402cbd36745" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="b4abacb7-01"
/dev/sr0: BLOCK_SIZE="2048" UUID="2025-06-03-11-44-48-84" LABEL="Ubuntu_GAs_7.1.10" TYPE="iso9660"
/dev/sda2: UUID="64b1cbeb-6657-4b8b-9332-defd0ce65725" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="b644dcb0-75fb-4229-81a7-71479e61b7a9"
/dev/sda1: PARTUUID="98e4368b-fed4-49ca-acf8-20aaa9ebba9d"
isak@UbuntuServer:~$

```

desmontar la partición backups

```
isak@UbuntuServer:~$ sudo umount /backups
isak@UbuntuServer:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0   50G  0 disk
├─sda1       8:1    0    1M  0 part
├─sda2       8:2    0   50G  0 part /
sdb          8:16   0   10G  0 disk
├─sdb1       8:17   0    5G  0 part /srv
└─sdb2       8:18   0    5G  0 part
sr0          11:0    1 58.5M  0 rom
isak@UbuntuServer:~$
```

crear el volumen físico

```
isak@UbuntuServer:~$ sudo pvcreate /dev/sdb2
WARNING: ext4 signature detected on /dev/sdb2 at offset 1080. Wipe it? [y/n]: y
Wiping ext4 signature on /dev/sdb2.
Physical volume "/dev/sdb2" successfully created.
isak@UbuntuServer:~$
```

crear el volumen group

```
isak@UbuntuServer:~$ sudo vgcreate vf_backups /dev/sdb2
Volume group "vf_backups" successfully created
isak@UbuntuServer:~$
```

Crear el Logical Volume (LV)

```
isak@UbuntuServer:~$ sudo lvcreate -L 3G -n lv_backups vf_backups
Logical volume "lv_backups" created.
isak@UbuntuServer:~$
```

formatear el LV

```
isak@UbuntuServer:~$ sudo mkfs.ext4 /dev/vf_backups/lv_backups
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 786432 4k blocks and 196608 inodes
Filesystem UUID: 4404e934-a66b-46e3-b2d0-798ee6337cf9
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

montar en backups

```
isak@UbuntuServer:~$ sudo mount /dev/vf_backups/lv_backups /backups
isak@UbuntuServer:~$
```

verificar el UUID

```
isak@UbuntuServer:~$ sudo blkid
/dev/sdb2: UUID="5W7U8D-MG9d-gc9N-Ftqs-Jfff-8Uoh-XX1J1n" TYPE="LVM2_member" PARTUUID="b4abacb7-02"
/dev/sdb1: LABEL="srv" UUID="07739997-1262-4aca-9d0a-9402cbd36745" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="b4abacb7-01"
/dev/sr0: BLOCK_SIZE="2048" UUID="2025-06-03-11-44-48-84" LABEL="UBox_GAs_7.1.10" TYPE="iso9660"
/dev/sda2: UUID="64b1cbeb-6657-4b8b-9332-de6d0ce65725" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="b644dcb0-75fb-4229-81a7-71479e61b7a9"
/dev/mapper/vf_backups-lv_backups: UUID="4404e934-a66b-46e3-b2d0-798ee6337cf9" BLOCK_SIZE="4096" TYPE="ext4"
/dev/sda1: PARTUUID="98e1368b-fed4-49ca-acf8-20aaa9ebba9d"
isak@UbuntuServer:~$
```

configurar montaje automático

```
GNU nano 8.7.1
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# <file system> <mount point> <type> <options>          <dump> <pass>
# / was on /dev/sda2 during curtin installation
/dev/disk/by-uuid/64b1cbeb-6657-4b8b-9332-de6d0ce65725 / ext4 defaults 0 1
UUID=4404e934-a66b-46e3-b240-798ee6337cf9 /backups ext4 defaults 0 2_
```

probar montaje

```
isak@UbuntuServer:~$ df -h
Filesystem                Size      Used Avail Use% Mounted on
tmpfs                     1.9G      1.1M    1.9G   1% /run
/dev/sda2                 49G       3.9G    43G   9% /
tmpfs                     4.7G       0      4.7G   0% /dev/shm
none                      1.0M       0      1.0M   0% /run/credentials/systemd-journald.service
tmpfs                     4.7G       0      4.7G   0% /tmp
none                      1.0M       0      1.0M   0% /run/credentials/systemd-resolved.service
none                      1.0M       0      1.0M   0% /run/credentials/systemd-networkd.service
none                      1.0M       0      1.0M   0% /run/credentials/getty@tty1.service
tmpfs                     943M      12K     943M   1% /run/user/1000
/dev/sdb1                 4.9G      1.3M    4.6G   1% /srv
/dev/mapper/uf_backups-lv_backups 2.9G     792K    2.8G   1% /backups
isak@UbuntuServer:~$
```

prueba de archivo "testfile"

```
isak@UbuntuServer:~$ sudo touch /backups/testfile
isak@UbuntuServer:~$ ls -l /backups
total 16
drwx----- 2 root root 16384 Jun 27 16:29 lost+found
-rw-r----- 1 root root      0 Jun 27 16:49 testfile
isak@UbuntuServer:~$
```